

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II
WEEK 2: LECTURE NOTE

CONTINUITY

Continuity at a Point and on an Open Interval

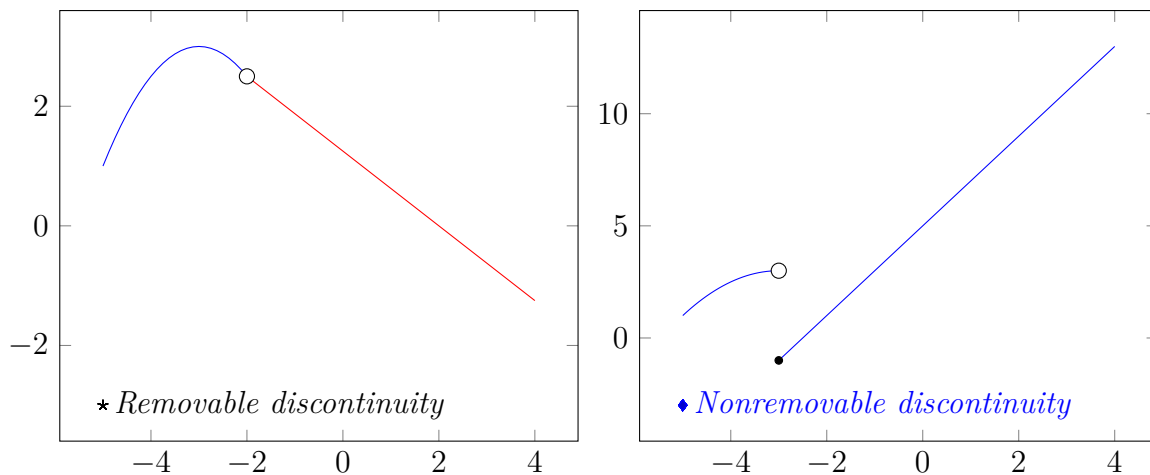
Definition 1. A function f is **continuous at c** when these three conditions are met.

1. $f(c)$ is defined.
2. $\lim_{x \rightarrow c} f(x)$ exists.
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

A function f is **continuous on an open interval** (a, b) when the function is continuous at each point in the interval. A function f that is continuous on the entire real number line $(-\infty, +\infty)$ is **everywhere continuous**.

If f fails to be continuous at c , then we say that f is **discontinuous (or not continuous)** at c .

Definition 2 (Removable and Nonremovable Discontinuity). A discontinuity at c is called **removable** when f can be made continuous by appropriately defining (or redefining) $f(c)$. Otherwise it is called a **nonremovable discontinuity**.



Example 3. Discuss the continuity of each function.

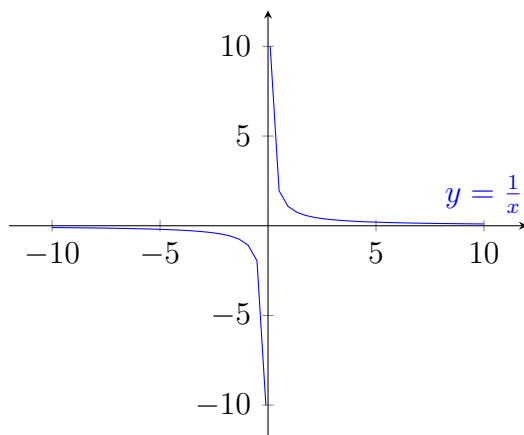
- (a) $f(x) = \frac{1}{x}$
- (b) $g(x) = \frac{x^2 - 1}{x - 1}$
- (c) $h(x) = \begin{cases} x + 1, & x \leq 0, \\ x^2 + 1, & x > 0 \end{cases}$
- (d) $y = \sin x$

Solution.

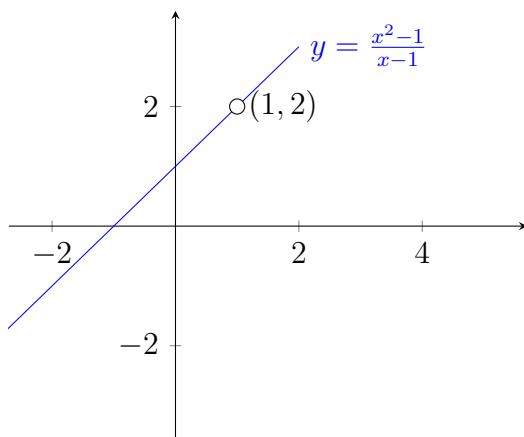
- (a) The domain of f is all nonzero real numbers. Since f is a rational function, for each $c \neq 0$,

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \frac{1}{x} = \frac{1}{c}.$$

Hence we can conclude that f is continuous at every nonzero c . We see from the graph of f given below, there is no way to define $f(0)$ so as to make the function f continuous at 0. Therefore, f has a **nonremovable discontinuity** at $x = 0$.



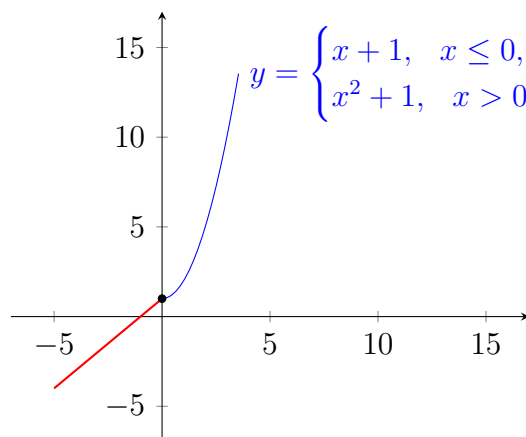
- (b) The domain of g is all real numbers except $x = 1$. Since g is a rational function, you can conclude that g is continuous at every x -value in its domain. By defining $g(1)$ as 2, the "redefined" function is continuous for all real numbers. Therefore, at $x = 1$ the function g has a **removable discontinuity**.



- (c) The domain of h is all real numbers. The function is continuous on $(-\infty, 0)$ and $(0, +\infty)$, and, because

$$\lim_{x \rightarrow 0} h(x) = 1 = h(0)$$

so that h is continuous at $x = 0$. Therefore, h is continuous on the entire real number line.



(d) The domain of y is all real numbers. It follows from limits of trigonometric functions that for each real number c

$$\lim_{x \rightarrow c} y(x) = \lim_{x \rightarrow c} \sin x = \sin c = y(c).$$

Therefore, the function y is continuous on its entire domain, $(-\infty, +\infty)$.

Continuity on a Closed Interval

Definition 4 (Definition of Continuity on a Closed Interval). A function f is continuous on the closed interval $[a, b]$ when f is continuous on the open interval (a, b) and

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x) = f(b).$$

The function f is **continuous from the right at a** and **continuous from the left at b** .

Example 5. Discuss the continuity of $f(x) = \sqrt{1 - x^2}$.

Solution. The domain of f is the closed interval $[-1, 1]$. It follows from the limit of radicals that f is continuous at every point in the open interval $(-1, 1)$. Since

$$\lim_{x \rightarrow -1^+} \sqrt{1 - x^2} = 0 = f(-1) \quad \text{and} \quad \lim_{x \rightarrow 1^-} \sqrt{1 - x^2} = 0 = f(1),$$

f is continuous from the right at -1 and continuous from the left at 1 . Therefore, f is continuous on the closed interval $[-1, 1]$.

Properties of Continuity

Theorem 6 (Properties of Continuity). If b is a real number and f and g are continuous at c , then the functions listed below are also continuous at c .

1. **Scalar multiple:** bf
2. **Sum or difference:** $f \pm g$
3. **Product:** fg
4. **Quotient:** $\frac{f}{g}$, $g(c) \neq 0$

Example 7. Each of the following functions is continuous at every point in its domain.

$$(a) \quad f(x) = x + \sin x \qquad (b) \quad g(x) = 3 \tan x \qquad (c) \quad h(x) = \frac{x^2 + 1}{\cos x}$$

Theorem 8 (Continuity of a Composite Function). *If g is continuous at c and f is continuous at $g(c)$, then the composite function given by $(f \circ g)(x) = f(g(x))$ is continuous at c .*

Example 9 (Testing for Continuity). *Describe the interval(s) on which each function is continuous.*

$$(a) \quad f(x) = \tan x \qquad (b) \quad g(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases} \qquad (c) \quad g(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Solution

- (a) The tangent function is undefined at $x = n\pi + \frac{\pi}{2}$, n is an integer. At all other points, f is continuous. So, $f(x) = \tan x$ is continuous on the open intervals

$$\dots, \left(-\frac{5\pi}{2}, -\frac{3\pi}{2}\right), \left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right), \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \frac{3\pi}{2}\right), \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right), \dots$$

- (b) Because $y = \frac{1}{x}$ is continuous except at $x = 0$ and the sine function is continuous for all real values of x , it follows from Continuity of Composite function that $y = \sin \frac{1}{x}$ is continuous at all real values except $x = 0$. At $x = 0$, the limit of f does not exist. So, g is continuous on the intervals $(-\infty, 0)$ and $(0, \infty)$.

- (c) This function is similar to the function in part (b) except that the oscillations are damped by the factor x . Using the Squeeze Theorem, you obtain

$$-|x| \leq x \sin \frac{1}{x} \leq |x|, \quad x \neq 0$$

and we can conclude that

$$\lim_{x \rightarrow 0} h(x) = 0.$$

So, h is continuous on the entire real number line.

Example 10. *If the function f given below is continuous on the entire real number line, then find a and b .*

$$f(x) = \begin{cases} x^2 - 3, & x < -1 \\ ax + 2, & -1 \leq x \leq 1 \\ x^3 - b, & x > 1 \end{cases}$$

Since f is continuous at $x = -1$ and $x = 1$, we have

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x) \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x).$$

So that

$$-a + 2 = -2 \quad \text{and} \quad a + 2 = 1 - b$$

Therefore, $a = 4$ and $b = -5$.

The Intermediate Value Theorem

One of important theorems concerning the behavior of functions that are continuous on a closed interval is the Intermediate Value Theorem.

Theorem 11 (Intermediate Value Theorem). *If f is continuous on the closed interval $[a, b]$, $f(a) \neq f(b)$, and k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a, b]$ such that*

$$f(c) = k.$$

The Intermediate Value Theorem often can be used to locate the **zeros of a function** that is continuous on a closed interval. Specifically, if f is continuous on the closed interval $[a, b]$, $f(a)$ and $f(b)$ differ in sign, then the Intermediate Value Theorem guarantees the existence of at least one zero of f in the closed interval $[a, b]$.

Example 12 (An Application of the Intermediate Value Theorem). *Use the Intermediate Value Theorem to show that the polynomial function*

$$f(x) = x^3 + 2x - 1$$

has a zero in the interval $[0, 1]$.

Solution. *Note that f is continuous on the closed interval $[0, 1]$. Because $f(0) = -1$ and $f(1) = 2$ it follows that $f(0) < 0$ and $f(1) > 0$. You can therefore apply the Intermediate Value Theorem to conclude that there must be some c in $[0, 1]$ such that*

$$f(c) = 0.$$

See the figure below.

